

BOOOOKSCORE: A SYSTEMATIC EXPLORATION OF BOOK-LENGTH SUMMARIZATION IN THE ERA OF LLMs

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ABSTRACT

Summarizing book-length documents ($>100K$ tokens) that exceed the context window size of large language models (LLMs) requires first breaking the input document into smaller chunks and then prompting an LLM to merge, update, and compress chunk-level summaries. Despite the complexity and importance of this task, it has yet to be meaningfully studied due to the challenges of evaluation: existing book-length summarization datasets (e.g., BookSum) are in the pretraining data of most public LLMs, and existing evaluation methods struggle to capture errors made by modern LLM summarizers. In this paper, we present the first study of the *coherence* of LLM-based book-length summarizers implemented via two prompting workflows: (1) *hierarchically merging* chunk-level summaries, and (2) *incrementally updating* a running summary. We obtain 1193 fine-grained human annotations on GPT-4 generated summaries of 100 recently-published books and identify eight common types of coherence errors made by LLMs. Because human evaluation is expensive and time-consuming, we develop an automatic metric, BOOOOKSCORE, that measures the proportion of sentences in a summary that do not contain any of the identified error types. BOOOOKSCORE has high agreement with human annotations and allows us to systematically evaluate the impact of many other critical parameters (e.g., chunk size, base LLM) while saving \$15K USD and 500 hours in human evaluation costs. We find that closed-source LLMs such as GPT-4 and Claude 2 produce summaries with higher BOOOOKSCORE than those generated by open-source models. While LLaMA 2 falls behind other models, Mixtral achieves performance on par with GPT-3.5-Turbo. Incremental updating yields lower BOOOOKSCORE but higher level of detail than hierarchical merging, a trade-off sometimes preferred by annotators. We release code and annotations to spur more principled research on book-length summarization.



github.com/lilakk/BooookScore

1 INTRODUCTION

Just two years ago, automatically-generated summaries were riddled with artifacts such as grammar errors, repetition, and hallucination (Zhao et al., 2020; Fabbri et al., 2020; Goyal & Durrett, 2021). Nowadays, such artifacts have mostly disappeared; in fact, Pu et al. (2023b) find that summaries generated by large language models (LLMs) are preferred over those written *by humans*, leading them to pronounce the death of summarization research. However, as with most prior work on summarization, the input documents in their study are relatively short ($<10K$ tokens). Widespread adoption of LLMs outside the research community has driven the development of a more ambitious task: summarizing *book-length* documents, which we define to be texts longer than 100K tokens.

As these documents exceed the context window limits of today’s LLMs (e.g., 8K tokens for GPT-4), summarizing them via prompt-based approaches necessitates heuristics to chunk the input, process each chunk, and then combine and compress the outputs (Wu et al., 2021).

Despite the promise that LLMs hold for long-context tasks, the research community still lacks a principled and systematic approach to evaluate their capabilities on book-length summarization. Our paper identifies three open challenges with evaluation: (1) *data contamination*, in which existing benchmarks such as BookSum (Kryscinski et al., 2022) are in the pretraining data of modern LLMs (Chang et al., 2023); (2) *an unexplored error distribution*, as most prior summarization research centers around short source documents and fails to capture coherence errors that are exacerbated by the “chunk and combine” book-length summarization setting; and (3) *a lack of any reliable automatic metric*, which requires careful design and validation against human annotations.

Contribution 1: A protocol for evaluating coherence in book-length summarization (§3). To mitigate the impact of data contamination, we design our evaluation framework around the use of *newly-published books*. We propose a reference-free protocol that leverages human annotation of the *coherence* of LLM-generated summaries (i.e., their logical connectedness) under different prompting strategies. Our protocol unifies and extends best-practices across disparate works in document understanding and evaluation research, including adoption of fine-grained annotation units, use of QA pairs to denote points of confusion, and a taxonomic breakdown of different coherence errors.

We validate our protocol by collecting 1193 span-level human annotations on GPT-4 generated summaries of a carefully curated set of 100 recently-published books (costing \$3K USD and 100 annotator hours) using two prompting strategies (hierarchical merging and incremental updating, shown in Figure 1). In categorizing these annotations into eight frequent error types, we reveal an error distribution in GPT-4 summaries that differs from that observed in prior studies on short-document summarizers; notably, we identify new error types (causal omissions, salience errors) through our book-length summarization setting (Table 1).

Contribution 2: An automatic metric—BOOOOKSCORE—to assess summary coherence (§4). Since our human evaluation is expensive, we follow recent work by developing an LLM-based evaluation metric called BOOOOKSCORE that identifies and explains instances of any of our eight established coherence errors in a given summary. Human validation shows that BOOOOKSCORE’s annotations are almost as reliable as those of human annotators, which allows us to automatically evaluate many other book-length summarization configurations. Because BOOOOKSCORE does not rely on gold summaries, it can easily be used to evaluate new LLM summarizers on any collection of newly-published books, ensuring that the metric will remain meaningful for LLMs of the future.

Contribution 3: A systematic evaluation of different LLMs using BOOOOKSCORE (§5). We use BOOOOKSCORE to evaluate the impact of several critical design decisions on the coherence of generated summaries, including the choice of prompting strategy, base LLM, and chunk size, a study that altogether cost \$10K (USD) in LLM API calls. Our findings include (1) hierarchical merging generally results in more coherent summaries but reduced level of detail compared to incremental updating; (2) GPT-4 and Claude 2 produce the most coherent summaries, while LLaMA 2 is substantially worse and fails to follow instructions; (3) increasing the chunk size does not improve hierarchical merging but does substantially benefit Claude 2 when using incremental updating; and (4) summary-level preference judgments are highly subjective and do not correlate with BOOOOKSCORE.

2 BACKGROUND: SUMMARIZING BOOK-LENGTH TEXTS WITH LLMs

Before discussing our evaluation protocol, we first outline two strategies—*hierarchical merging* and *incremental updating*—for prompting an LLM to summarize book-length documents that exceed its maximum context size. In both strategies, the length of the input document necessitates first dividing it into smaller chunks and then repeatedly merging, updating, and/or compressing chunk-level partial summaries (Figure 1). While neither strategy is well-explored by published research, hierarchical merging essentially adapts the strategy proposed by Wu et al. (2021) to zero-shot prompting, while incremental updating resembles chain-of-density prompting proposed for short-document summarization (Adams et al., 2023). Both are implemented in widely-used open-source LLM libraries such as LangChain,¹ but the relative merits of each method remain unexplored.

¹LangChain implements incremental updating via refine and hierarchical merging via map-reduce.

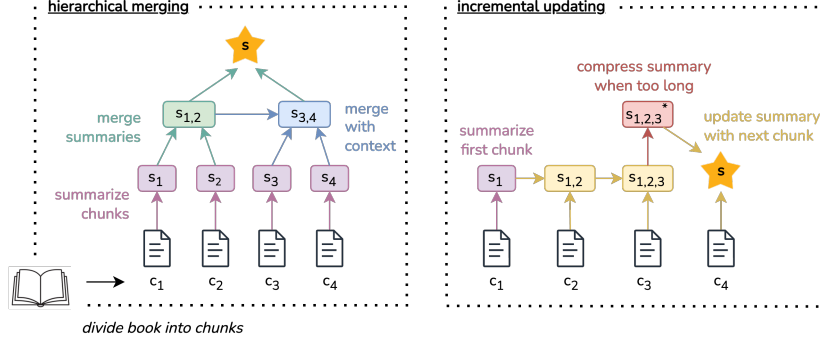


Figure 1: To perform book-length summarization, we first divide a book into smaller chunks that fit within the context window of an LLM. Then, we explore two strategies for summarization: (1) *hierarchical merging*, in which chunks are first summarized and then the corresponding summaries merged via separate prompts; and (2) *incremental updating*, in which a global summary is updated and compressed as we step through the book chunk-by-chunk.

More specifically, both strategies assume an LLM with context window size W is used to summarize an input document D whose length $L \gg W$. We thus split D into non-overlapping chunks $c_1, c_2, \dots, c_{\lceil \frac{L}{C} \rceil}$ where $C < W$ is the length of each chunk.²

Hierarchical merging: Wu et al. (2021) propose a method in which an LLM (in their case, GPT-3) is fine-tuned via reinforcement learning to summarize each chunk and then hierarchically merge the chunk-level summaries until one summary is left of the entire input document. This method has since been simplified into a zero-shot prompting strategy without further training, as shown in Figure 1 (left). Hierarchical merging requires three unique prompts for (1) summarizing an input chunk, (2) merging chunk-level summaries, and (3) merging summaries with added context from previously-generated merged summaries. We ensure that the total length of each prompt and its associated inputs is less than $W - G_l$, where G_l is a hyperparameter controlling summary length that varies depending on the level l . Summaries are recursively merged until only one summary (of the full book) remains; see Appendix A.1 for further details.

Incremental updating: It is possible that since hierarchical merging necessitates summarizing portions of the input document without complete context, it may introduce more coherence errors. For example, in the first level, chunks towards the end of the book will be summarized without knowledge of what came before, which can lead to incoherent summaries especially for non-linear or multi-perspective narratives. We thus explore an alternate prompting strategy—*incremental updating* (Figure 1, right)—that iterates through each chunk in order while continuously updating a global summary with salient information. While this method may be better able to handle inter-chunk dependencies than hierarchical merging, it requires more complicated prompts for (1) summarizing an input chunk, (2) updating the global summary $s_{1,2,\dots,i-1}$ with information from the current chunk c_i , and (3) compressing the global summary when it exceeds the maximum summary length G_n . See Appendix A.2 for a full specification of incremental updating.

3 EVALUATING COHERENCE OF BOOK SUMMARIES

In this section, we define our framework for human evaluation of coherence errors in book-length summarization. Our framework involves: (1) corpus collection focusing on newly-published books, (2) unification and extension of best-practices from prior document understanding and evaluation literature to guide data annotation, and (3) analysis of human annotations centered around emergent coherence error categories of summaries generated by modern LLMs.

²We ensure each chunk ends at a sentence boundary.